Transfer: thermal energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose an equation or diagram for energy transfer. Explain when it stops working.

SUCCESS CRITERIA - I can...

1. Distinguish warming, phase change and work.
2. Use correct signs and units.
3. Explain how a predicted result depends on what you treat as true.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “internal energy” to explain your answer. Sentence starters are optional.

phase change: a change between states such as solid and liquid

internal energy: energy stored in a system’s microscopic motion/interactions

efficiency: useful output divided by input

01 Warm-up: bring back an earlier idea

A heat engine absorbs Q_{hot}=**2400 J** and rejects Q_{cold}=**1680 J** each cycle. **Find work** and efficiency.

Word help: cycle = a process that returns a system to its initial state

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q = mc\,\Delta T$; $Q = mL$; $\Delta U = Q - W_{by}$; $\eta = W_{out}/Q_{hot}$. **Choose an equation** that fits and use the supplied constants. **State how you handle heat** loss, phase changes, and a full cycle.

Worked reasoning: A system takes **120 J** of heat, does **50 J** of work and increases internal energy by **70 J**. An energy account must include transfers and storage rather than treating all heat as useful work.

Your turn: Draw its system boundary, input, output and storage change.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A hot block and cool water are isolated together. **Draw initial heat flow** and **describe equilibrium** without inventing a final temperature.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Hot block

Cool water

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Transfer: thermal energy

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A beaker contains **0.10 kg** of ice at **0°C**. Melt it and warm the water to **10°C**.
1. Use L_f = **334,000 J/kg** and c_{water} = **4200 J**/(kg·°C).
2. **Show both stages** in the energy account and **find the total energy required**.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

An engine absorbs **500 J** and produces **150 J** work in a cycle. **Find rejected heat** and **explain why an energy balance** alone does not establish that any temperature pair can achieve that efficiency.

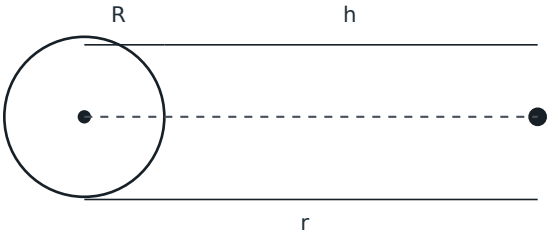
Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simulated source has GM=**288 m³/s²**. At r=**4 m**, find escape speed and **name two things your calculation** treats as true.
Label the distances or energy levels used in your reasoning.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: 1 2 3

07 Exit check

A heat engine between **500 K** and **300 K** absorbs **1000 J**.
1. **Find ideal maximum work** and **explain why actual work usually** falls below it.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Word help: ideal = using stated simplifying conditions

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **0.5 kg** sample with c=**400 J**/(kg·K) warms **2 K** while doing **60 J** of work on surroundings.
1. **Find heat input under** $\Delta U = mc\,\Delta T$ and **name one thing** that must be true for that equation to apply.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3